

How to install GNU make on windows

Installing GNU Make on Windows can be done through several methods, with using a package manager like Chocolatey or Scoop being the easiest.

Here are the most common methods to install make on Windows:

Method 1: Using Chocolatey (Recommended)

If you have Chocolatey installed, this is the fastest method.

1. Open **PowerShell** or **Command Prompt** as Administrator.
2. Run the following command:

```
cmd
```

```
choco install make
```

3. Restart your terminal to ensure the command is recognized.

Method 2: Using Scoop

If you prefer Scoop, this method is also efficient.

1. Open **PowerShell**.
2. Run the following commands:

```
powershell
```

```
scoop bucket add main  
scoop install make
```

```
scoop bucket add main  
scoop install make
```

Method 3: Using Winget (Built-in Windows Package Manager)

Windows 10/11 comes with winget, which can install GnuWin32 Make.

1. Open **Command Prompt** or **PowerShell**.
2. Run:

```
cmd
```

```
winget install GnuWin32.Make
```

```
winget install GnuWin32.Make
```

3. After installation, you may need to add the installation path to your system PATH (usually C:\Program Files (x86)\GnuWin32\bin).

Method 4: MinGW (For Developers)

If you are developing C/C++ projects, you likely need a compiler, and MinGW includes make.

1. Install MinGW (or MSYS2) and ensure you have checked the option for make or mingw32-make in the installer.
2. Add the bin directory of your MinGW installation to your system PATH (e.g., C:\MinGW\bin).
3. *Optional:* To use make instead of mingw32-make, navigate to the bin folder and rename mingw32-make.exe to make.exe.

Method 5: Manual Installation (GnuWin32)

1. Download the **Setup** binary from the GnuWin32 Make page.
2. Run the installer and note the installation folder (e.g., C:\Program Files (x86)\GnuWin32).
3. Add the bin folder (e.g., C:\Program Files (x86)\GnuWin32\bin) to your Windows PATH environment variable.

Verifying Installation

Open a new terminal (CMD or PowerShell) and type:

```
cmd
```

```
make --version
```

make --version

If successful, you will see the version of GNU Make installed.

Using Make in Git Bash

If you use Git Bash, you might need to ensure make is properly added to your ~/.bashrc or ~/.bash_profile if it does not work automatically after the above steps.

Alternatively, downloading the make.exe from ezwinports and copying it to the bin folder of your Git installation folder works as well.